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Negative pressure suction-based fixing device

Abstract

Disclosed is a negative pressure suction-based fixing device comprising a housing, a suction and holding member, a negative pressure creating assembly and a sealing member. The housing has an accommodating cavity thereinside. The suction and holding member has an abutting and holding surface recessed to form a suction cavity communicated with the accommodating cavity. The abutting and holding surface has a mounting structure connected to the sealing member, so that the sealing member is arranged on the abutting and holding surface for abutting against the suction surface. The sealing member can be elastically deformed to fit with the suction surface so as form an enclosed space together with the cooperation of the suction cavity and the suction surface. In this way, it is difficult for the negative pressure suction-based fixing device to fall off from a rough surface.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) The present application is a Continuation Application of PCT Application No. PCT/CN2024/095031 filed on May 23, 2024, the contents of which are incorporated herein by reference in their entirety.

TECHNICAL FIELD
(1) The present disclosure relates to suction and holding apparatus, in particular to negative pressure suction-based fixing devices.
BACKGROUND OF THE INVENTION
(2) A negative pressure suction-based fixing device is a type of equipment that allows a suction and holding member to be abutted and held on a suction surface such as a surface of a vehicle body, and generates negative pressure thereinside, allowing the suction and holding member to be firmly attached to the suction surface. Since the suction and holding member relies on internal negative pressure to generate suction force, there is a high requirement for the airtightness of the space formed by the cooperation between the suction and holding member and the suction surface,

because the suction and holding member may fall off from the suction surface if air enters the space.
(3) However, existing negative pressure suction-based fixing devices can only be suctioned onto smooth surfaces. When these devices are suctioned onto rough surfaces, it becomes difficult to form the enclosed space due to the unevenness of the rough surfaces, which often leads to detachment of the fixing devices from the rough surfaces.

SUMMARY OF THE INVENTION

(4) The present disclosure mainly provides negative pressure suction-based fixing devices, solving the above problem that existing negative pressure suction-based fixing devices are prone to detachment when suctioned onto rough surfaces.
(5) To address the aforementioned technical problem, the present disclosure provides a negative pressure suction-based fixing device, which may include a housing, a suction and holding member, a negative pressure creating assembly and a sealing member. The interior of the housing may comprise an accommodating cavity. The suction and holding member may have an abutting and holding surface that is recessed to form a suction cavity, which is in communication with the accommodating cavity. The abutting and holding surface may be provided with a mounting structure connected to the sealing member, such that the sealing member is arranged on the abutting and holding surface. The sealing member may be configured to be abutted and held on a suction surface, and may be capable of being elastically deformed to fit with the suction surface to form an enclosed space together with the suction cavity and the suction surface. The negative pressure creating assembly may be configured to generate negative pressure in the accommodating cavity so as to create negative pressure in the enclosed space.
(6) In some embodiments, the sealing member is a colloid.
(7) In some embodiments, the mounting structure includes a receiving groove, and the sealing member is disposed in the receiving groove.
(8) In some embodiments, the receiving groove has a ring shape, surrounding the outer circumference of the suction cavity; and the sealing member has a ring shape.
(9) In some embodiments, the negative pressure suction-based fixing device includes a release film. The release film may be covered over the side of the sealing member that is away from the housing to protect the sealing member.
(10) In some embodiments, the negative pressure suction-based fixing device includes a solar panel and a battery. The solar panel may be disposed on the housing, with a light absorbing surface at least partially exposed through the housing, and may be configured to convert solar energy into electrical energy. The battery may be configured to supply power to the negative pressure creating assembly and may be capable of storing the electrical energy converted by the solar panel.
(11) In some embodiments, the negative pressure suction-based fixing device includes a charging interface that is configured to connect to an external power supply to the supply power to the battery.
(12) In some embodiments, the negative pressure suction-based fixing device comprises a first charging circuit, a second charging circuit and a comparator. The first charging circuit may include a first switching unit and may be configured to connect to the solar panel. The second charging circuit may include a second switching unit and may be configured to connect to the external power supply. One end of the first switching unit may be connected to the solar panel, and another end of the first switching unit may be connected to the battery. One end of the second switching unit may be connected to the external power supply, and another end of the second switching unit may be connected to the battery. The comparator may have a first input end connected to the first charging circuit, a second input end connected to the second charging circuit, and an output end configured to output a comparison result. The first switching unit may be closed or opened in response to the comparison result from the comparator, and the second switching unit may be closed or opened in response to the comparison result from the comparator.

(13) In some embodiments, the first charging circuit further includes a first protection unit and the second charging circuit further includes a second protection unit. One end of the first protection unit is connected to the first switching unit, and the other end is connected to the battery. Similarly, one end of the second protection unit is connected to the second switching unit, and the other end is connected to the battery. Both the first protection unit and the second protection unit may be configured to prevent reverse current.
(14) In some embodiments, the negative pressure suction-based fixing device further includes a magnetic member that may be disposed on the housing and may be configured to magnetically attract a part to be secured.
(15) In some embodiments, the magnetic member can rotate relative to the housing, allowing it to move closer to or farther away from the housing during rotation, so as to adjust the angle between the magnetic member and the housing.
(16) In some embodiments, the housing has opposite first and second ends along its axial direction, with the suction and holding member disposed at the first end, and the magnetic member disposed at the second end.
(17) In some embodiments, the negative pressure suction-based fixing device further includes a rotating member rotatably connected to the housing via a rotating shaft, with the magnetic member fixedly disposed on it.
(18) In some embodiments, a slot is provided on a side of the rotating member that is away from the housing, where at least part of the magnetic member is disposed.
(19) In some embodiments, the negative pressure creating assembly includes: a cylinder barrel having a receiving cavity thereinside that is communicated with the accommodating cavity, a cylinder piston arranged in the receiving cavity, and a driving element configured to drive the cylinder piston to move in a reciprocating manner, thereby generating negative pressure in the accommodating cavity.
(20) In some embodiments, the negative pressure suction-based fixing device further includes an activation assembly including a first movable part and a switch part, wherein an end of the first movable part may be arranged outside the accommodating cavity and another end of the first movable part may be arranged in the accommodating cavity, and the first movable part may be capable of being moved towards the interior of the accommodating cavity to actuate the switch part, thereby activating the negative pressure creating assembly.
(21) In some embodiments, the direction of movement of the first movable part is perpendicular to the abutting and holding surface; and when the suction and holding member is suctioned onto the suction surface, the first movable part can be abutted against the suction surface, thereby suffering pressure from the suction surface to move towards the interior of the accommodating cavity.
(22) In some embodiments, the switch part is a bi-directional switch, and the first movable part can further be moved towards the exterior of the accommodating cavity to actuate the bi-directional switch, thereby activating the negative pressure creating assembly.
(23) In some embodiments, the negative pressure suction-based fixing device further includes an air inlet assembly configured to communicate the accommodating cavity with an exterior of the housing, allowing the suction and holding member to be detached from the suction surface.
(24) In some embodiments, the air inlet assembly includes a second movable part with an end arranged outside the housing and another end arranged inside the housing; and when subjected to pressure, the second movable part can open an air channel between the accommodating cavity and the exterior of the housing.
(25) As previously mentioned, the negative pressure suction-based fixing device provided in this application includes a housing with an accommodating cavity inside; a suction and holding member having an abutting and holding surface recessed to form the suction cavity in communication with the accommodating cavity and provided with a mounting structure; a sealing member connected to the sealing member and arranged on the abutting and holding surface,

capable of elastically deforming to fit the suction surface and forming an enclosed space in cooperation with the suction cavity and the suction surface; and a negative pressure creating assembly configured to generate negative pressure in the accommodating cavity to create negative pressure in the enclosed space. In this way, due to the fact that the sealing member with strong deformability is arranged on the abutting and holding surface of the suction and holding member, when the negative pressure suction-based fixing device is abutted against a rough suction surface, the sealing member can be deformed into an uneven shape matching the uneven contour of the suction surface to fit with the suction surface to enhance the airtightness of the space formed by the suction and holding member, the sealing member and the suction surface, thus making it difficult for the negative pressure suction-based fixing device to fall off when suctioned onto the rough surface.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a schematic structural diagram of a negative pressure suction-based fixing device in some embodiments of the present disclosure from a perspective;
(2) FIG. **2** is a schematic structural diagram of a negative pressure suction-based fixing device in some embodiments of the present disclosure from another perspective;
(3) FIG. **3** is a schematic exploded diagram of the structure shown in FIG. **1**;
(4) FIG. **4** is a schematic section view of a suction and holding member and a sealing member in some embodiments of the present disclosure;
(5) FIG. **5** is a schematic structural diagram of a negative pressure suction-based fixing device in other embodiments of the present disclosure;
(6) FIG. **6** is a schematic structural diagram of a negative pressure suction-based fixing device without a suction and holding member in some embodiments of the present disclosure;
(7) FIG. **7** is a block diagram of a charging circuit of a negative pressure suction-based fixing device in some embodiments of the present disclosure;
(8) FIG. **8** is a block diagram of a connection structure for a comparator in some embodiments of the present disclosure;
(9) FIG. **9** is a schematic structural diagram of a magnetic member rotated in some embodiments of the present disclosure;
(10) FIG. **10** is a schematic exploded diagram of the structure shown in FIG. **9**;
(11) FIG. **11** is a schematic structural diagram of a lower housing, a negative pressure creating assembly, an activation assembly and an air inlet assembly in some embodiments of the present disclosure;
(12) FIG. **12** is a schematic structural diagram of a negative pressure creating assembly, an activation assembly and an air inlet assembly in some embodiments of the present disclosure; and
(13) FIG. **13** is a schematic structural diagram of a negative pressure creating assembly (in an exploded view), an activation assembly and an air inlet assembly in some embodiments of the present disclosure.
DETAILED DESCRIPTION
(14) The present disclosure is further described in detail below through specific embodiments in combination with the drawings, wherein similar elements across different embodiments adopt associated similar element labels. In the following embodiments, many details are described to facilitate better understanding of the application. However, those skilled in the art can readily recognize that some features can be omitted in different cases or can be replaced by other elements, materials and methods. In some cases, some operations pertinent to this disclosure are not explicitly shown or described in the specification to prevent the core aspects of this disclosure from

being obscured by excessive detail, and detailing these operations is not deemed necessary, as those skilled in the art can fully comprehend them based on the specification and the general technical knowledge in the field.

(15) Furthermore, the characteristics, operations or features described in the specification may be combined in any suitable manner to form various embodiments. At the same time, the steps or actions described in the method may be sequenced, adjusted, or rearranged in a manner apparent to those skilled in the art. Therefore, the sequences in the specification and the drawings are merely intended for the clear description of an embodiment and do not necessarily represent a required sequence, unless otherwise specified as a mandatory sequence.

(16) The serial numbers, such as “first”, “second”, etc., assigned to the parts in this document are only used to distinguish the described objects and do not have any sequential or technical meaning. The terms “connect” and “couple” used in this document, unless specifically stated otherwise, refer to both direct and indirect connections (couplings).

(17) The present disclosure provides a negative pressure suction-based fixing device that can be suctioned onto a suction surface. This device may be provided with a connecting part, allowing a part to be secured to be secured onto the device, thus achieving fixation of the part to be secured on the suction surface. The connection between the connecting part and the part to be secured can include, but is not limited to, buckle connection, magnetic snap connection, and interference fit connection. In an exemplary application, the device may be used in a vehicle-mounted environment; where it can be secured on a vehicle body, enabling an electronic device to be fixed onto the device by the connecting part, so that the content displayed by the electronic device can be viewed by users. Of course, the negative pressure suction-based fixing device provided in the present disclosure is not limited to the vehicle-mounted environments, and it can be used in any context that requires fixation of a part to be secured.

(18) Please refer to FIGS. **1-5**. The negative pressure suction-based fixing device may include a housing **10**, a suction and holding member **20**, a negative pressure creating assembly **30** and a sealing member **40**. The housing **10** may have an accommodating cavity **11** inside. Specifically, the housing **10** may include an upper housing and a lower housing. These two housings may be detachably connected and together form the accommodating cavity **11**.

(19) The suction and holding member **20** may be designed to be suctioned onto a suction surface. It can be disposed on the housing **10**, for instance, on an end face of the housing **10** along its axial direction or on an annular side wall of the housing **10**. As shown in FIG. **2**, an end of the suction and holding member **20** away from the housing **10** may have an abutting and holding surface **21**. The abutting and holding surface **21** may be recessed to form a suction cavity **22** that may be in communication with the accommodating cavity **11**; that is, the internal gas pressure in the suction cavity **22** may be the same as that in the accommodating cavity **11**. Examples of the suction and holding member **20** may include a suction cup or a suction ring. To prevent air leakage from an enclosed space between the suction and holding member **20** and the suction surface, the material of the suction and holding member **20** may be made of a flexible material. The flexible material may be deformable, allowing for better attachment of the suction and holding member **20** to the suction surface.

(20) The abutting and holding surface may be provided with a mounting structure designed to secure the sealing member **40**. The sealing member **40** may be connected to the mounting structure so as to be arranged on the abutting and holding surface **21**. The sealing member **40** may be designed to abut against the suction surface and can be elastically deformed to conform to the surface of the suction surface. This allows the suction cavity **22**, the sealing member **40** and the suction surface to cooperate to form the enclosed space. For example, in some embodiments, the sealing member **40** may be a colloid, such as a semi-solid, soft colloid like a gel. Compared to harder colloids such as rubber, semi-solid and soft colloids exhibit stronger elastic deformation capabilities and can more effectively adapt to the uneven shapes of suction surfaces.

(21) The negative pressure creating assembly **30** may be designed to generate negative pressure within the accommodating cavity **11**. Since the accommodating cavity **11** is in communication with the enclosed space, the negative pressure creating assembly **30** can create negative pressure in the enclosed space, allowing the suction and holding member **20** to generate a suction and holding effect due to negative pressure. The negative pressure creating assembly **30** can be powered on to stably generate negative pressure in the enclosed space, which is more stable and reliable than natural suction methods, such as by squeezing out the air in the enclosed space to make the suction and holding member **20** attach to the suction surface.
(22) The negative pressure suction-based fixing device may be provided with the sealing member **40** with strong deformability on the abutting and holding surface **21** of the suction and holding member **20**. Accordingly, when the negative pressure suction-based fixing device is abutted against a rough suction surface, the sealing member **40** can deform to match the uneven contour of the rough suction surface, ensuring a secure fit with the surface. This allows the suction and holding member **20** to enhance the airtightness of the space formed between it and the suction surface via the sealing member **40**, making it difficult for the negative pressure suction-based fixing device to detach from the rough surface.
(23) As shown in FIGS. **2** and **4**, in some embodiments, the mounting structure includes a receiving groove **211**, and the sealing member **40** is disposed within this groove. In this way, the connection between the sealing member **40** and the suction and holding member **20** is more reliable, preventing the sealing member **40** from falling off the suction and holding member **20**. Specifically, the thickness of the sealing member **40** may be slightly greater than the depth of the receiving groove **211**, so that when the negative pressure suction-based fixing device is abutted against the suction surface, the suction surface may be abutted against the sealing member **40**, resulting in a better sealing effect for the enclosed space compared with the suction and holding member **20** abutted against the suction surface.
(24) In some embodiments, the receiving groove **211** may be annular and arranged around the outer circumference of the suction cavity **22**, and the sealing member **40** may also be designed in an annular shape. This annular shape of the sealing member **40** enables the outer circumference of the suction cavity **22** to achieve good sealing with the suction surface, thereby enhancing the sealing performance of the enclosed space. The annular shape may be, for example, a circular ring, a square ring, or any other irregularly shaped ring, as long as it is enclosed.
(25) Referring to FIG. **5**, in some embodiments, the negative pressure suction-based fixing device may also include a release film **50** that is covered over the side of the sealing member **40** away from the housing **10** to protect the sealing member **40**. When in the as-shipped condition, the release film **50** can protect the sealing member **40** during transportation and storage, preventing dust from sticking to the surface of the sealing member **40**. If the surface of the sealing member **40** adheres to too many solid particles stick to, it may lead to a reduction in the sealing performance of the enclosed space, resulting in poor suction of the negative pressure suction-based fixing device.
(26) Referring to FIG. **3**, the negative pressure suction-based fixing device may also include a solar panel **60** and a battery. The solar panel **60** may be a component capable of converting solar energy into electrical energy. The solar panel **60** may be disposed on the housing **10** and may have a light absorbing surface **61**. At least a portion of the light absorbing surface **61** may be exposed through the housing **10**. Preferably, to enhance the energy conversion efficiency of the solar panel **60**, the entire light absorbing surface **61** may be exposed through the housing **10**, allowing the solar panel **60** to fully harness solar energy. The battery may be configured to supply power to the negative pressure creating assembly **30** and may be capable of storing the electrical energy converted by the solar panel **60**. With the solar panel **60** disposed on the housing **10**, during daytime driving, the solar panel **60** can convert solar energy into electrical energy and store it into the battery. This means that during daytime, the solar panel **60** can continuously supply power to the battery, enabling the battery to continuously supply power to the negative pressure creating assembly **30**.

This improves the battery life of the negative pressure suction-based fixing device, reducing the need for frequent power charging and enhancing user experience. Even when not driving during daytime, the solar panel **60** can convert solar energy into electrical energy, compensating for the natural discharge of the battery during periods of non-use, ensuring that the battery of the negative pressure suction-based fixing device can also have sufficient power supply at night.
(27) As shown in FIG. **3** and FIG. **6**, in some embodiments, the housing **10** may have a first end face **121** and a second end face **131** that are axially opposite to each other on the housing **10**. The suction and holding member **20** may be disposed on the first end face **121** of the housing **10**. The wall of the suction cavity **22** may be provided with a first through-hole **221**, and the first end face **121** of the housing **10** may be provided with a second through-hole **1211**. The suction cavity **22** may be in communication with the accommodating cavity **11** via the first through-hole **221** and the second through-hole **1211**. The solar panel **60** may be disposed on the second end face **131** of the housing **10**. Since the suction and holding member **20** is disposed on the first end face **121** of the housing **10**, which typically faces the vehicle body, while the second end face **131** of the housing **10** faces the vehicle window, disposing the solar panel **60** on the second end face **131** of the housing **10** is advantageous for the solar panel **60** to absorb solar energy through the window.
(28) As shown in FIG. **3**, in some embodiments, the negative pressure suction-based fixing device further includes a charging interface **70** that is configured to connect to an external power supply to supply power to the battery. Even though the solar panel **60** can be used to charge the battery, the battery's stored energy may be depleted during long periods of travel at night. Therefore, in some embodiments, the negative pressure suction-based fixing device may incorporate two charging modes: charging with the solar panel **60** and charging with the external power supply, enhancing the reliability of the negative pressure suction-based fixing device during nighttime or cloudy/rainy conditions.
(29) Further, in some embodiments, referring to FIG. **7** and FIG. **8**, the negative pressure suction-based fixing device may also include a first charging circuit **81**, a second charging circuit **82** and a comparator **83**. The first charging circuit **81**, which may include a first switching unit **811**, may be designed to connect to the solar panel **60**. The second charging circuit **82**, which may include a second switching unit **821**, may be designed to connect to the external power supply.
(30) One end of the first switching unit **811** may be connected to the solar panel **60**, and another end of the first switching unit **811** may be connected to the battery. One end of the second switching unit **821** may be connected to the external power supply, and another end of the second switching unit **821** may be connected to the battery. The first input end of the comparator **83** may be connected to the first charging circuit **81**, and the second input end of the comparator **83** may be connected to the second charging circuit **82**. The output end of the comparator **83** may be configured to output a comparison result that is may be, for example, the comparison of the charging current between the first charging circuit **81** and the second charging circuit **82**. The first switching unit **811** may respond to the comparison result from the comparator **83** by switching on or off, and similarly, the second switching unit **821** may also respond to the comparison result by switching on or off.
(31) For example, when the solar panel **60** charges the battery, if the external power supply is not connected and the first charging circuit **81** is greater than that of the second charging circuit **82**, the comparator **83** may output a first comparison result. In response to this, the first switching unit **811** is in the on state and the second switching unit **821** is in the off state. When the external power supply is connected, and the charging current of the second charging circuit **82** is greater than that of the first charging circuit **81**, the comparator **83** may output a second comparison result. In response to this, the first switching unit **811** is in the off state and the second switching unit **821** is in the on state. This means that when the external power supply is connected, it ensures that the second charging circuit **82**, powered by the external power supply, is connected, while the first charging circuit **81**, powered by the solar panel **60**, is disconnected. This prevents both charging

modes from operating simultaneously.
(32) Furthermore, in some embodiments, the first charging circuit **81** may also comprise a first protection unit **812**, and the second charging circuit **82** may also comprise a second protection unit **822**. One end of the first protection unit **812** may be connected to the first switching unit **811**, and another end may be connected to the battery. One end of the second protection unit **822** may be connected to the second switching unit **821**, and another end may be connected to the battery. These protection units **812**, **822** may be designed to prevent reverse current. The first protection unit **812** and the second protection unit **822** may be, for example, diodes.
(33) In some embodiments, as shown in FIGS. **1-3**, the negative pressure suction-based fixing device may also comprise a magnetic member **91** disposed on the housing **10**. Preferably, the magnetic member **91** may be disposed on an end of the housing **10** away from the first end face **121**. In some examples, the magnetic member **91** may be disposed on the side of the solar panel **60** away from the housing **10**, and may expose at least a portion of the light absorbing surface **61**. For example, the magnetic member may be designed in an annular or curved shape.
(34) The magnetic member **91** may be configured to magnetically attract a part to be secured. Specifically, in some embodiments, the connecting part may be configured as the magnetic member **91**. For instance, in applications involving the securing of electronic equipment, the suction and holding member **20** may be used to be suctioned and supported on the vehicle body, and the magnetic member **91** may be used to magnetically attract the electronic equipment. Of course, the magnetic member **91** may be used to magnetically attract the vehicle body, while the suction and holding member **20** may be used to suction and support the electronic equipment. Generally, the housing of the electronic equipment may be made of metal material which can magnetically attract the magnetic member **91**. Alternatively, the interior of the electronic equipment may be provided with a metal or magnetic part that can magnetically attach to the magnetic member **91**. In this respect, leveraging the typical characteristics of the electronic equipment enables a reliable connection between the electronic equipment and the negative pressure suction-based fixing device.
(35) Additionally, in some embodiments, the negative pressure suction-based fixing device may also include a wireless charging coil that can charge an electronic equipment wirelessly when the magnetic member **91** is magnetically attracted by the electronic equipment.
(36) In some embodiments, as shown in FIG. **9** and FIG. **10**, the magnetic member **91** can be rotated relative to the housing **10** to allow the magnetic member **91** to move closer and farther away from the housing **10**, so as to adjust an angle between the magnetic member **91** and the housing **10**. Since the housing **10** is secured to the suction surface by the suction and holding member **20**, and the housing **10** is relatively fixed in position with respect to the suction surface, the rotation of the magnetic member **91** relative to the housing **10** is equivalent to the rotation of the magnetic member **91** relative to the suction surface, and this rotation can adjust the angle between the magnetic member **91** and the suction surface. In an vehicle-mounted environment, when the housing **10** is secured to the vehicle body by the suction and holding member **20**, the magnetic member **91** may be driven to rotate relative to the housing **10** to adjust an angle of an electronic device secured to the magnetic member **91** relative to a driving seat, so that an appropriate angle of the display screen of the electronic device facing towards to a user can be chosen by the user. In this way, the angle of the electronic device can be adjusted without needing to detach the negative pressure suction-based fixing device, providing convenience to the user. Moreover, the rotation angle of the magnetic member **91** relative to the housing **10** is large, allowing the angle of the electronic device relative to the user to be not overly restricted by the structure of the vehicle body, which can meet the needs for adjusting the angle of the part to be secured relative to the driving seat within a wide range.
(37) Accordingly, with the negative pressure suction-based fixing device presented in the present disclosure, the magnetic member **91** is capable of magnetically attracting the part to be secured by

mounting the magnetic member **91** on the housing **10**, allowing the negative pressure suction-based fixing device to be detachably connected to the part to be secured. Moreover, the magnetic member **91** can be magnetically connected to the part to be secured, whether it is on the side close to the housing or on the side far away from the housing; and the part to be secured can be rotated at any angle along the contact surface where magnetic attraction occurs, making the relative position of the negative pressure suction-based fixing device and the part to be secured change more variable. Furthermore, by designing the magnetic member **91** to be rotated relative to the housing **10** to adjust the angle between the magnetic member **91** and the housing **10**, the angle of the part to be secured, which is connected to the magnetic member **91**, relative to the suction surface can be adjusted without removing the negative pressure suction-based fixing device, thus meeting the needs for the user to conveniently adjust the angle of the part to be secured relative to the suction surface in a wide range.

(38) Please refer to FIG. **9** and FIG. **10**, in some embodiments, the housing **10** may have a first end **12** and a second end **13** that are opposite to each other along the axial direction of the housing **10**. The suction and holding member **20** may be disposed on the first end **12** of the housing **10**, and the magnetic member **91** may be disposed on the second end **13** of the housing **10**. The first end **12** may include a first end face **121** and a part of the annular side wall of the housing **10** close to the first end face **121**, and the second end **13** may include a second end face **131** and a part of the annular side wall of the housing **10** close to the second end face **131**. By disposing the suction and holding member **20** and the magnetic member **91** at both ends of the housing **10**, in general, the suction and holding member **20** may be suctioned and supported on the suction surface, the location of the magnetic member **91** may be further away from the suction surface, the magnetic member **91** is less likely to be interfered by the suction surface when rotating, thus allowing for a larger rotation angle of the magnetic member **91**.

(39) The magnetic member **91** may rotate relative to the housing **10** along a rotating shaft **93**, and the axial direction of the rotating shaft **93** may be perpendicular to the axial direction of the housing **10**. In some embodiments, the housing **10** may be approximately cylindrical in shape, with the axial direction of the rotating shaft being perpendicular to the axial direction of the cylinder. The rotation angle α between the magnetic member **91** and the housing **10** may range from more than 0 to 270 degrees, i.e. $0<\alpha\leq 270$ degrees. In some embodiments, the second end face **131** of the housing **10** may be configured as a plane, and when the magnetic member **91** is in an initial state, the magnetic member **91** may be arranged in a stacked manner on the second end face **131**, the angle between a face of the magnetic member **91** close to the second end face **131** and the second end face **131** is 0 degrees; and when the magnetic member **91** starts rotating from the initial state, the magnetic member **91** may be rotated in a direction away from the second end face **131**.

(40) As shown in FIG. **9** and FIG. **10**, in some embodiments, the negative pressure suction-based fixing device may also include a rotating member **92** that is rotatably connected to the housing **10** by the rotating shaft **93**. The magnetic member **91** may be fixedly disposed on the rotating member **92**. Since it is difficult to modify the structure of the magnetic member **91** to install the rotating shaft **93**, in some embodiments, the magnetic member **91** may be secured to the rotating member **92**, so that the magnetic member **91** can be rotatably connected to the housing **10**. The magnetic member **91** may be disposed on the rotating member in a non-removable connection or a removable connection. In some embodiments, the side of the rotating member **92** away from the housing **10** may be provided with a slot, and at least a portion of the magnetic member **91** may be disposed in the slot. A cover may be arranged on the slot and may serve as a protective cover to protect the magnetic member **91** inside the rotating member **92**. In other embodiments, the magnetic member **91** may be embedded within the rotating member **92** or attached to the surface of the rotating member **92** using methods such as adhesive.

(41) As shown in FIGS. **11-13**, in some embodiments, the negative pressure creating assembly **30** may include a driving element **31**, a cylinder piston **32** and a cylinder barrel **33**. Inside the cylinder

barrel **33**, there may be a receiving cavity that is communicated with the accommodating cavity **11** and accommodates the cylinder piston **32**. The driving element **31** may be configured to drive the cylinder piston **32** to reciprocate, which generates negative pressure within the accommodating cavity **11**. Since the negative pressure creating assembly **30** is driven by the driving element **31**, this allows the user to control the operation of the negative pressure creating assembly **30** by controlling the starting and stopping of the driving element **31**.

(42) The driving element **31** may be a motor. The output shaft of the motor may drive the cylinder piston **32** to move through a transmission mechanism, for example, driving the cylinder piston **32** to reciprocate via a crank mechanism.

(43) As shown in FIGS. **11-13**, in some embodiments, the negative pressure suction-based fixing device may also include an activation assembly **111** that includes a first movable part **1111** and a switch part **1112**. An end of the first movable part **1111** may be arranged outside the accommodating cavity **11**, and another end of the first movable part **1111** may be arranged inside the accommodating cavity **11**. The first movable part **1111** may be capable of moving towards the interior of the accommodating cavity **11** to actuate the switch part **1112** to activate the negative pressure creating assembly **30**. This allows the user to operate the first movable part **1111** from the exterior of the housing **10** to activate the negative pressure creating assembly **30**, facilitating easy operation of the first movable part **1111** by the user, and also allowing the user to decide when to perform negative pressure suction.

(44) In some embodiments, the direction of movement of the first movable part **1111** may be perpendicular to the abutting and holding surface **21**. When the suction and holding member **20** is suctioned onto the suction surface, the first movable part **1111** can be abutted against the suction surface to be moved towards the interior of the accommodating cavity **11** under pressure from the suction surface, thus activating the negative pressure creating assembly **30**. In this way, when the suction and holding member **20** is pressed onto the suction surface by the user, the first movable part **1111** can be activated on its own, enabling the negative pressure creating assembly **30** to operate simultaneously with the user's pressing action. There is no need for the user to manipulate the first movable part **1111**, thereby simplifying the activation process of the negative pressure creating assembly **30**. Preferably, the portion of the first movable part **1111** located outside the housing **10** is arranged inside the suction cavity **22** of the suction and holding member **20**. Of course, in other embodiments, the portion of the first movable part **1111** located outside the housing **10** may be arranged outside the suction cavity **22**.

(45) In some embodiments, the switch part **1112** may be a bi-directional switch, and the first movable part **1111** can also move towards the exterior of the accommodating cavity **11**, actuating the bi-directional switch so as to activate the negative pressure creating assembly **30**. As the suction and holding member **20** may loosen during suction and support, when it loosens, the negative pressure suction-based fixing device may need to be pressed again, which is troublesome. With the provision of the bi-directional switch, the negative pressure creating assembly **30** may be activated when the first movable part **1111** is pressed or released, thereby preventing the loosening of the suction and holding member **20** and allowing the suction and holding member **20** to be more stably suctioned onto the suction surface.

(46) As shown in FIGS. **11-13**, in some embodiments, the negative pressure suction-based fixing device may also include an air inlet assembly **112** configured to communicate the accommodating cavity **11** with the exterior of the housing **10**, enabling the suction and holding member **20** to be separated from the suction surface. With the provision of the air inlet assembly **112**, the user can actively and smoothly separate the suction and holding member **20** from the suction surface. For example, the air inlet assembly **112** may include a second movable part **1121** and a blocking part. An end of the second movable part **1121** may be arranged outside the housing **10**, and another end of the second movable part **1121** may be arranged inside the housing **10**. The second movable part **1121**, when subjected to pressure, may enable the blocking part to move to communicate the air

channel between the accommodating cavity **11** and the exterior of the housing **10**.
(47) In other embodiments, the negative pressure suction-based fixing device can also allow to press a button, which disconnects the circuit of the negative pressure creating assembly **30**, causing it to stop working, thus enabling the user to remove the suction and holding member **20** from the suction surface.
(48) The present application has been illustrated above with reference to specific embodiments, which are merely provided for the purpose of understanding the present application and are not intended to limit the scope of the present application. For those skilled in the art, it is possible to make certain simple derivations, modifications, or substitutions based on the principles of the present application.

Claims

1. A negative pressure suction-based fixing device, comprising a housing having an accommodating cavity thereinside; a suction and holding member having an abutting and holding surface that is recessed to form a suction cavity in communication with the accommodating cavity and is provided with a mounting structure; a sealing member that is connected to the mounting structure to be arranged on the abutting and holding surface, wherein the sealing member is configured to be abutted and held on a suction surface, and is capable of being elastically deformed to fit with the suction surface so as to form an enclosed space cooperatively with the suction cavity and the suction surface, wherein the sealing member is a colloid; and a negative pressure creating assembly configured to generate negative pressure in the accommodating cavity so as to create negative pressure in the enclosed space.
2. The negative pressure suction-based fixing device according to claim 1, wherein the mounting structure includes a receiving groove, and the sealing member is mounted in the receiving groove.
3. The negative pressure suction-based fixing device according to claim 2, wherein the receiving groove is ring-shaped and arranged around an outer circumference of the suction cavity; and the sealing member is ring-shaped.
4. The negative pressure suction-based fixing device according to claim 1, further comprising a release film that is covered over a side of the sealing member away from the housing to protect the sealing member.
5. The negative pressure suction-based fixing device according to claim 1, further comprising: a solar panel mounted on the housing and configured to convert solar energy into electrical energy, with a light absorbing surface that is at least partially exposed through the housing; and a battery configured to supply power to the negative pressure creating assembly and be capable of storing the electrical energy converted by the solar panel.
6. The negative pressure suction-based fixing device according to claim 5, further comprising a charging interface configured to connect to an external power supply to supply power to the battery.
7. The negative pressure suction-based fixing device according to claim 6, further comprising: a first charging circuit configured to connect to the solar panel and having a first switching unit that has one end connected to the solar panel and another end connected to the battery; a second charging circuit configured to connect to the external power supply and having a second switching unit that has one end connected to the external power supply and another end connected to the battery; and a comparator having a first input end connected to the first charging circuit, a second input end connected to the second charging circuit, and an output end configured to output a comparison result, such that the first switching unit and the second switching unit are each responsive to the comparison result from the comparator to be in an on or off state.
8. The negative pressure suction-based fixing device according to claim 7, wherein the first charging circuit further comprises a first protection unit configured to prevent reverse current and

having one end connected to the first switching unit and another end connected to the battery; and the second charging circuit further comprises a second protection unit configured to prevent reverse current and having one end connected to the second switching unit and another end connected to the battery.

9. The negative pressure suction-based fixing device according to claim 1, further comprising a magnetic member mounted on the housing and configured to magnetically attract a part to be secured.

10. The negative pressure suction-based fixing device according to claim 9, wherein the magnetic member is capable of rotating relative to the housing to move closer to or farther away from the housing, so as to adjust an angle between the magnetic member and the housing.

11. The negative pressure suction-based fixing device according to claim 10, wherein the housing has opposite first and second ends along its axial direction, with the suction and holding member mounted at the first end of the housing and the magnetic member at the second end of the housing.

12. The negative pressure suction-based fixing device according to claim 10, further comprising a rotating member rotatably connected to the housing via a rotating shaft, with the magnetic member fixedly mounted thereon.

13. The negative pressure suction-based fixing device according to claim 12, wherein a side of the rotating member away from the housing is provided with a slot, and at least a portion of the magnetic member is mounted in the slot.

14. The negative pressure suction-based fixing device according to claim 1, the negative pressure creating assembly comprises: a cylinder barrel having inside a receiving cavity in communication with the accommodating cavity; a cylinder piston arranged in the receiving cavity; and a driving element configured to drive the cylinder piston to reciprocate so as to generate negative pressure within the accommodating cavity.

15. The negative pressure suction-based fixing device according to claim 1, further comprising an activation assembly including: a first movable part having one end arranged outside the accommodating cavity and another end arranged in the accommodating cavity, and being capable of moving towards an interior of the accommodating cavity; and a switch part configured to be actuated by movement of the first movable part towards the interior of the accommodating cavity so as to activate the negative pressure creating assembly.

16. The negative pressure suction-based fixing device according to claim 15, wherein a direction of the movement of the first movable part is perpendicular to the abutting and holding surface; and when the suction and holding member is suctioned onto the suction surface, the first movable part is capable of abutting against the suction surface, thereby suffering pressure from the suction surface to move towards the interior of the accommodating cavity.

17. The negative pressure suction-based fixing device according to claim 15, wherein the switch part is a bi-directional switch; and the first movable part is further capable of being moved towards an exterior of the accommodating cavity to actuate the bi-directional switch so as to activate the negative pressure creating assembly.

18. The negative pressure suction-based fixing device according to claim 1, further comprising an air inlet assembly configured to communicate the accommodating cavity with an exterior of the housing, allowing the suction and holding member to be detached from the suction surface.

19. The negative pressure suction-based fixing device according to claim 18, wherein the air inlet assembly comprises a second movable part having one end disposed outside the housing and another end disposed inside the housing, capable of opening an air channel between the accommodating cavity and the exterior of the housing when subjected to pressure.

20. A negative pressure suction-based fixing device, comprising a housing having an accommodating cavity thereinside; a suction and holding member having an abutting and holding surface that is recessed to form a suction cavity in communication with the accommodating cavity and is provided with a mounting structure; a sealing member that is connected to the mounting

structure to be arranged on the abutting and holding surface, wherein the sealing member is configured to be abutted and held on a suction surface, and is capable of being elastically deformed to fit with the suction surface so as to form an enclosed space cooperatively with the suction cavity and the suction surface; a negative pressure creating assembly configured to generate negative pressure in the accommodating cavity so as to create negative pressure in the enclosed space; and a release film that is covered over a side of the sealing member away from the housing to protect the sealing member.